Question	Answer	Marks	Partial Marks	Guidance
20(b)	$3(\sqrt{7}-2)$ oe final answer	**3**	**B2** for $\frac{9(2+\sqrt{7})}{4-7}$ or better or **M1** for $\frac{9}{2+\sqrt{7}} \times \frac{2-\sqrt{7}}{2-\sqrt{7}}$ oe	Accept e.g. $-3(2-\sqrt{7})$
21(a)	(0, –3)	**1**		
21(b)	[y =] $-\frac{1}{2}x+12\frac{1}{2}$ final answer	**5**	**M1** for [grad AB =] $\frac{15-7}{5-1}$ oe **M1** for –1 divided by *their* grad of AB **B1** for [midpoint =] (3, 11) **M1** for substituting *their* (3, 11) into y = (*their m*)x + c	Accept –0.5 and 12.5 or $\frac{25}{2}$ Correct answer in different form or unsimplified implies 4 marks *their m* ≠ *their* gradient AB e.g. $y - 11 = -\frac{1}{2}(x-3)$
22(a)	(–1, 1)	**5**	**B2** for $3x^2+2x-1$ or **B1** for two terms correct **M1** for setting *their* dy/dx =0 **M1** for $(3x-1)(x+1)$ FT *their* 3 term quadratic **A1** for $x = -1$ [and $x = \frac{1}{3}$]	For B marks isw e.g. 2nd derivative found Accept e.g. formula, complete the square

Question	Answer	Marks	Partial Marks	Guidance
22(b)	Correct shape of positive cubic with $\left(\frac{1}{3}, -\frac{5}{27}\right)$ in correct quadrant AND $\left(\frac{1}{3}, -\frac{5}{27}\right)$ is a minimum	**2**	**B1** for correct shape of positive cubic	At STM consider: Does the curve have to go through (0,0)? Does it have to have the max in the correct quadrant?
22(c)	$k \geq 1$ and $k \leq -\frac{5}{27}$	**2**	**FT** *their* '$y = 1$' from (a) **B1FT** for each or **SC1FT** for $k > 1$ and $k < -\frac{5}{27}$	
23(a)	$\frac{x^3}{8}$	**2**	**B1** for $\frac{x^3}{k}$ or $\frac{x^k}{8}$	
23(b)	6	**4**	**B3** for $4x - x = 2(x + 3)$ oe OR **B1** for 2^{-x} or 2^{4x} **B1** for $2^{2(x+3)}$	e.g. $x - 0.25x = ½ (x + 3)$ Use of consistent other bases is acceptable e.g. $4^{-0.5x}$ or $16^{-0.25x}$ or 4^{2x} e.g. $16^{0.5(x+3)}$